

CN Capstone Project

Mac 2 - Edge / Reverse Proxy / Load Balancer Documentation

Phase 1 / Detailed Evidence Record

Team: Packet Forge Domain: cn-capstone.test Machine: Mac 2 - cn-edge

1. Scope of This Mac 2 Document

Mac 2 is the HTTPS edge machine. This report documents only the Mac 2 work and evidence: nginx configuration, TLS termination, public HTTPS verification, reverse proxying, A/B load balancing, cache-control headers, and the backend failure/recovery demonstration. Wireshark evidence is deliberately excluded because the packet captures were performed on other project machines.

2. Correct Project Topology

Machine	Project role / hostname	IPv4	Interface	Mask / gateway	Service
Mac 1	cn-dns - Private DNS + client	10.7.9.245	en0	255.255.224.0 / 10.7.0.1	dnsmasq :53
Mac 2	cn-edge - HTTPS edge / reverse proxy / LB	10.7.5.53	en0	255.255.224.0 / 10.7.0.1	nginx HTTPS :8443
Mac 3	Backend A	10.7.22.10	en0	255.255.224.0 / 10.7.0.1	HTTP :3001
Mac 4	Backend B + client	10.7.7.25	en0	255.255.224.0 / 10.7.0.1	HTTP :3002

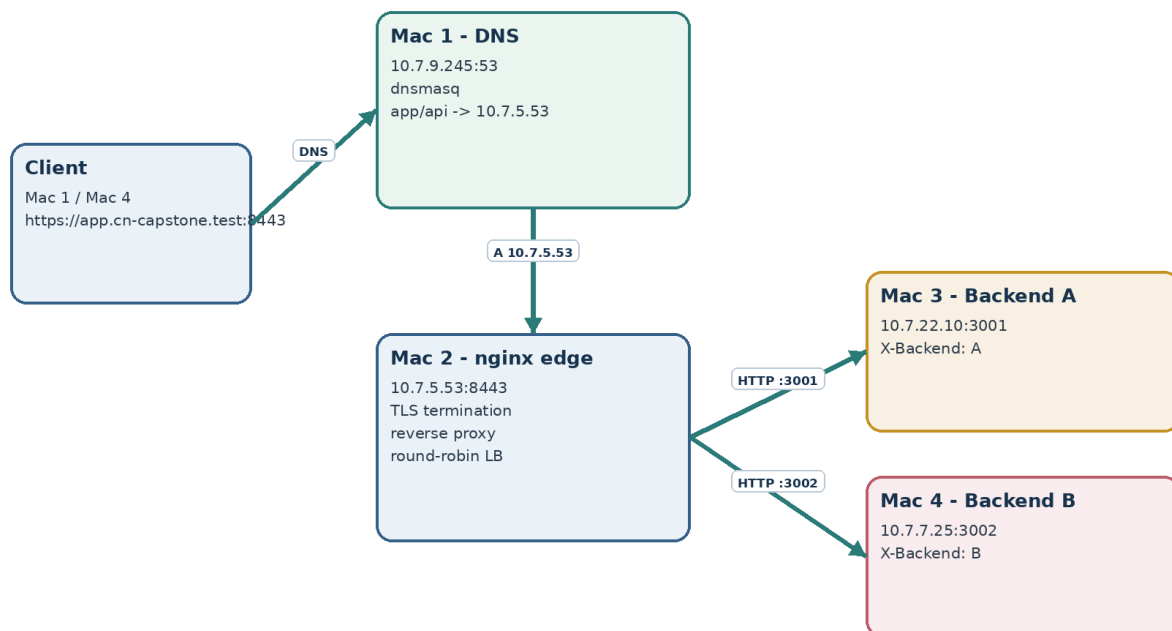
Mac 2 network identity: hostname cn-edge, interface en0, IPv4 10.7.5.53, mask 255.255.224.0 (/19), gateway 10.7.0.1, DNS 10.7.9.245.

3. Mac 2 Responsibilities

- Single HTTPS entry point: <https://app.cn-capstone.test:8443>.
- TLS termination with the project certificate.
- Reverse proxy from nginx to Backend A and Backend B.
- Round-robin load balancing across 10.7.22.10:3001 and 10.7.7.25:3002.
- Pass-through of backend caching headers and backend identity header.
- Failure demonstration by stopping Backend A and observing continued public service via Backend B.

4. Project Topology Diagram

Phase 1 project service topology - Mac 2 view



Mac 2 is the only documented HTTPS edge. Wireshark evidence is intentionally excluded from this Mac 2 report.

Figure 1. Corrected and spaced service-topology diagram. The client, DNS, Mac 2 edge, and two backend branches are kept visually separate.

5. nginx Configuration

```
● ● ● nginx upstream + HTTPS server

upstream backend_pool {
    server 10.7.22.10:3001;
    server 10.7.7.25:3002;
}

server {
    listen 8443 ssl;
    server_name app.cn-capstone.test;

    ssl_certificate /opt/homebrew/etc/nginx/certs/server.crt;
    ssl_certificate_key /opt/homebrew/etc/nginx/certs/server.key;

    location / {
        proxy_pass http://backend_pool;
        proxy_http_version 1.1;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto https;
        proxy_set_header Connection "";
    }
}
```

Figure 2. Supplied Mac 2 nginx configuration showing the upstream pool, HTTPS server on 8443, certificate paths, and proxy headers.

6. nginx Validation

● ● ● nginx verification commands

```
nginx -t
sudo nginx -T 2>&1 | grep -A5 -B2 "upstream backend_pool"
sudo lsof -nP -iTCP:8443 -sTCP:LISTEN
```

Figure 3. Supplied Mac 2 validation evidence. `nginx -t` succeeded and the HTTPS listener was observed on TCP 8443.

7. TLS Termination and Public HTTPS

HTTPS and TLS termination on Mac 2



Client-to-edge traffic is HTTPS; nginx terminates TLS before forwarding internally over HTTP.

Figure 4. Spaced TLS flow diagram. Client traffic is HTTPS/TLS to Mac 2; nginx terminates TLS and makes the internal HTTP proxy request.

● ● ● final HTTPS curl evidence

```
curl -v https://app.cn-capstone.test:8443

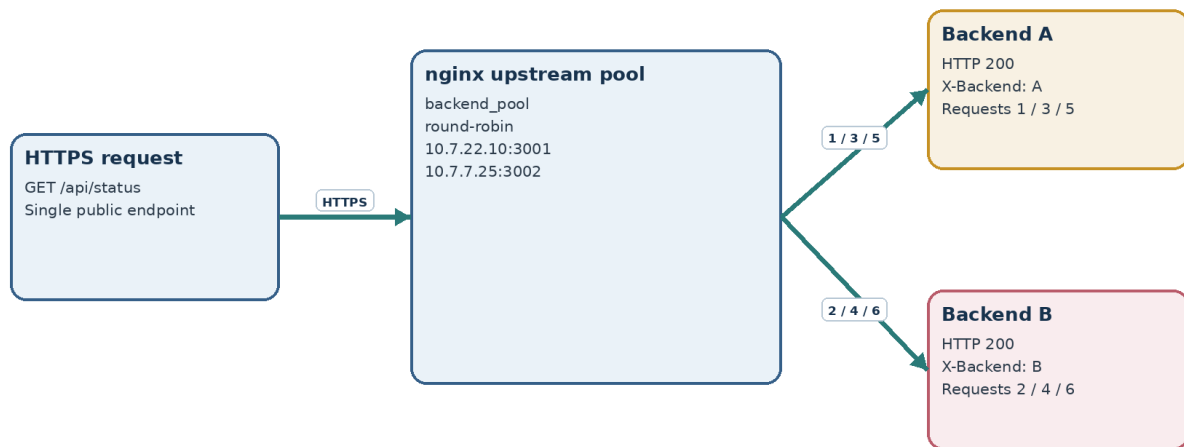
Connected to app.cn-capstone.test (10.7.5.53) port 8443
SSL connection using TLSv1.3 / AEAD-CHACHA20-POLY1305-SHA256
subject: ... CN=app.cn-capstone.test
subjectAltName: host "app.cn-capstone.test" matched cert
SSL certificate verify ok.

< HTTP/1.1 200 OK
< Server: nginx/1.31.6
< Cache-Control: max-age=60
< ETag: "B-v1"
< X-Backend: B
```

Figure 5. Supplied Mac 2 `curl -v` evidence. DNS resolves to 10.7.5.53, TLS 1.3 succeeds, the certificate subject/SAN matches, verification succeeds without `-k`, and HTTP 200 is returned.

8. Reverse Proxy and Load Balancing

Round-robin load balancing



Observed sequence: A, B, A, B, A, B.

Figure 6. Spaced round-robin diagram with distinct branches for Backend A and Backend B.

```
● ● ● six-request load balancing proof

REQUEST 1
HTTP/1.1 200 OK
X-Backend: A

REQUEST 2
HTTP/1.1 200 OK
X-Backend: B

REQUEST 3
HTTP/1.1 200 OK
X-Backend: A

REQUEST 4
HTTP/1.1 200 OK
X-Backend: B

REQUEST 5
HTTP/1.1 200 OK
X-Backend: A

REQUEST 6
HTTP/1.1 200 OK
X-Backend: B
```

Figure 7. Supplied six-request proof. HTTP 200 is returned for all six requests and X-Backend alternates A, B, A, B, A, B.

Configured upstream pool

10.7.22.10:3001 = Backend A; 10.7.7.25:3002 = Backend B. The public client never needs to address these ports directly.

9. HTTP Caching and Conditional Validation

```
●●● HTTP caching response headers

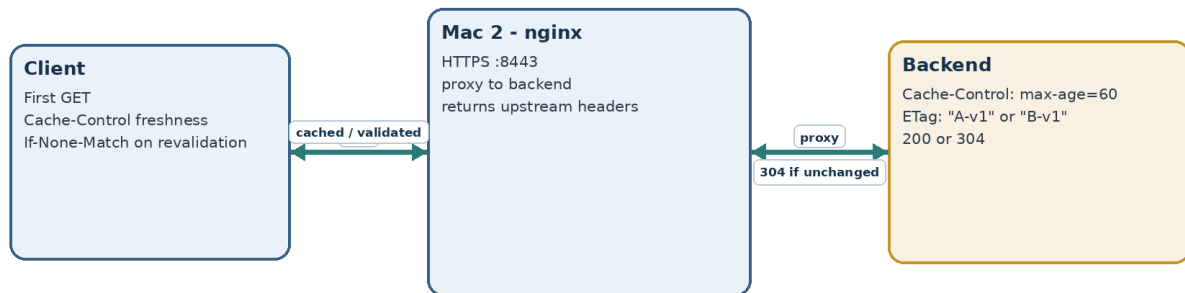
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Thu, 01 Oct 2026 10:26:03 GMT
Content-Type: application/json
Connection: keep-alive
Cache-Control: max-age=60
ETag: "B-v1"
X-Backend: B
```

Figure 8. Supplied Mac 2 public response headers: Cache-Control: max-age=60, ETag: "B-v1", and X-Backend: B.

Cache-Control: max-age=60 means the response is fresh for 60 seconds. The client may reuse the representation during that freshness interval before it needs revalidation.

ETag: the version token identifies the representation. A client can send If-None-Match; if the representation is unchanged, the backend can return 304 Not Modified.

Cache-Control and ETag validation



Cache-Control: max-age=60 defines freshness; ETag enables conditional validation and 304 Not Modified.

Figure 9. Spaced cache-validation diagram. The client revalidates through nginx and can receive a 304 response when the ETag is unchanged.

10. Backend Failure Demonstration and Recovery

The selected scenario is Option A: stop Backend A on Mac 3. The affected layer is the backend/application service on TCP port 3001. DNS on Mac 1 and nginx on Mac 2 remain part of the public path.

```
● ● ● failure demo - before

===== BEFORE FAILURE =====
Request 3 -> X-Backend: A
Request 4 -> X-Backend: B
Request 5 -> X-Backend: A
Request 6 -> X-Backend: B
```

Figure 10. Before failure: the public path had normal A/B participation.

```
● ● ● failure demo - after

===== AFTER BACKEND A FAILURE =====
Request 1 -> X-Backend: B
Request 2 -> X-Backend: B
Request 3 -> X-Backend: B
Request 4 -> X-Backend: B
Request 5 -> X-Backend: B
Request 6 -> X-Backend: B
```

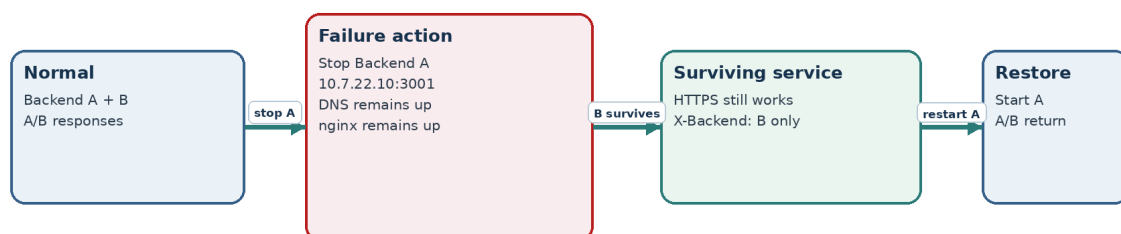
Figure 11. During failure: requests continued through Backend B.

```
● ● ● failure demo - restored

===== SYSTEM RESTORED =====
Request 1 -> X-Backend: A
Request 2 -> X-Backend: A
Request 3 -> X-Backend: B
Request 4 -> X-Backend: A
Request 5 -> X-Backend: B
Request 6 -> X-Backend: A
```

Figure 12. After restoration: both A and B appeared again.

Backend failure and recovery



Failure is demonstrated at the backend/application layer, not at DNS or the nginx edge.

Figure 13. Spaced failure/recovery diagram with distinct normal, failure, survival, and restore stages.

11. Evidence Allocation Across the Team

Project area	Primary machine(s)	Mac 2 status
Private DNS / dnsmasq	Mac 1	Referenced as dependency; not a Mac 2 configuration task
Backend A / port 3001	Mac 3	Upstream dependency; tested through nginx
Backend B / port 3002	Mac 4	Upstream dependency; tested through nginx
HTTPS edge / TLS / nginx	Mac 2	Primary Mac 2 evidence
A/B load balancing	Mac 2	Primary Mac 2 evidence
Caching headers / 304 behavior	Backend machines, observed through Mac 2 public path	Mac 2 public-path evidence
Wireshark Task G	Captured on other project machines	Not included in Mac 2 report

12. Mac 2 Final Evidence Checklist

Evidence item	What it proves	Status
Project topology and Mac 2 identity	10.7.5.53 / en0 / gateway / DNS	Verified
nginx upstream + server block	Correct backends, TLS listener, proxy_pass	Verified
nginx -t / port 8443	Syntax valid and edge listening	Verified
curl -v without -k	DNS, TCP, TLS, certificate verification, HTTP 200	Verified
Six public requests	A/B/A/B/A/B round-robin behavior	Verified
Cache headers	Cache-Control, ETag, X-Backend	Verified
Failure / recovery	B-only service during A failure, A/B after restore	Recorded
Wireshark	Not a Mac 2 task in this document	Intentionally omitted

13. Mac 2 Final Status

- Mac 2 is the single HTTPS entry point at 10.7.5.53:8443.
- TLS 1.3 and certificate verification succeed without the -k bypass.
- nginx forwards requests to Backend A and Backend B.
- Six requests produced the observed A/B/A/B/A/B sequence.
- Public responses expose Cache-Control and ETag headers.
- Stopping Backend A leaves the public service available through Backend B, and restoring Backend A returns the two-backend system.
- No Wireshark claim is made for Mac 2; the packet-level section is intentionally outside this Mac 2 report.